

Ashley Miller

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SUMMARY

Biomedical and mechanical engineer (MS Biomedical Engineering, Columbia; BS Engineering Physics, Cornell) working across the full development cycle — from CAD and 3D-printed prototypes to FEA, DOE-driven optimization, and cGMP verification & validation. Hands-on experience building medical devices, automated diagnostic platforms, and AI-assisted hardware systems, with a track record of owning projects end-to-end and translating engineering work into field- and clinic-ready outcomes.

EDUCATION

Columbia University

MS in Biomedical Engineering, GPA: 4.0/4.0

New York, NY

Sep 2024 — Feb 2026

Coursework: Drug & Gene Delivery, Quantitative Physiology, Biomechanics, Numerical Methods, Robotics, Life Sciences Entrepreneurship

Cornell University

BS in Engineering Physics, GPA: 3.5/4.0

Ithaca, NY

Aug 2020 — May 2024

Coursework: Mathematical Physics, Mechanics, E&M, Quantum Mechanics, Statistical Thermodynamics, Electronic Circuits, Nuclear Engineering, OOP & Data Structures, MATLAB, Cell & Developmental Biology, Orthopedic Tissue Mechanics

TECHNICAL SKILLS

Design & Fabrication: SolidWorks, Autodesk Inventor, Fusion 360, AutoCAD, Engineering Drawings, DFM, GD&T, 3D Printing, Manual Mill/Lathe, Soldering, Laser Cutting, Embedded Systems

Simulation & FEA: Ansys (Mechanical, HFSS, Fluent), Genesys, OnScale, LTSpice, PyBullet

Programming: Python, C++, MATLAB, Java, ROS

Data & Statistics: Excel, Minitab, DOE, statistical analysis

Lab & Regulatory: cGMP, V&V protocols, wet lab, DOE, root cause analysis

PROFESSIONAL EXPERIENCE

Brekland

Mechanical Engineering Intern (part-time)

Brooklyn, NY

Feb 2026 — May 2026

- Owned full-cycle development of two AI-assisted vision systems for precision sprayers, delivering a field-ready MVP in 3 months.
- Managed the BOM, designed components in Fusion 360, and programmed C++/Arduino controls; conceived a novel object-selection algorithm and built Python GUI tools for calibration and monitoring.
- Engineered and 3D-printed custom mounting hardware and a benchtop validation enclosure for structured, repeatable testing under real-world conditions; authored operator documentation and a market-informed commercialization roadmap.

Werfen

R&D Systems Engineering Co-op (full-time)

Norcross, GA

Jun 2025 — Aug 2025

- Executed cGMP-compliant V&V protocols on two novel automated blood-testing platforms; debugged 5+ hardware/software failures through hands-on repairs and cross-functional work with internal teams and external manufacturers.
- Reduced reagent waste by 64% by optimizing pipettor parameters through data-driven DOE and assay-constraint analysis.
- Investigated well spillage via root-cause analysis — modeling centrifuge ramp profiles in Excel and analyzing packaging, storage, and lot variation in Minitab — exposing an injection-molding defect in specific batches.

Adie Research Group, Cornell University

Undergraduate Researcher (part-time)

Ithaca, NY

Jun 2023 — May 2024

- Innovated mechanical components enabling a 65 MHz increase in ultrasound frequency for an ARF-OCE (Acoustic Radiation Force – Optical Coherence Elastography) imaging system, incorporating user feedback to accelerate clinical translation.
- Led FEA in Ansys to reconstruct phantom mechanical properties, predict acoustic-radiation-force effects, and validate against gold-standard equipment — bridging simulation and experimental biomechanics.

TTM Technologies

RF and Microwave Engineering Intern (full-time)

Syracuse, NY

Jun 2023 — Aug 2023

- Evaluated RF component designs end-to-end: modeled manufacturability via tolerance analysis in HFSS/AutoCAD, ran hands-on testing with network analyzers, and benchmarked results against customer specifications.
- Diagnosed the root cause of a manufacturing defect by x-ray imaging the internal structure, reverse-engineering it in simulation, and validating with experimental test data.

Cornell Electric Vehicles

Ithaca, NY

Drivetrain Sub-system Lead (part-time)

Jan 2023 — May 2023

- Led 20 engineers through the team's first full jiggling and assembly project, synchronizing all mechanical subsystems with tolerance-driven jig design; partnered with five leads on data-driven motor selection (torque, RPM, performance).

Drivetrain Engineer (part-time)

Oct 2020 — Jan 2023

- Designed and manufactured drivetrain components (sprockets, brake mount, axles) using Autodesk Inventor, Ansys stress analysis, and manual machining; modeled holistic vehicle efficiency with physics-based calculations.

Climate Engineering Lab, Cornell University

Ithaca, NY

Undergraduate Research Assistant (part-time)

Oct 2021 — Jan 2022

- Edited climate-model code in MATLAB and Fortran on the GAUSS project (with Dr. Douglas MacMartin) to simulate cooling of the Great Barrier Reef; generated IPCC-style plots of regional climate change.

TEACHING

Private Tutor

Mountain Lakes, NJ

Self-Employed (part-time)

Sep 2018 — March 2020 & Feb 2026 -- Present

- Tutored students from grades 2 through 11, specializing in STEM subjects including honors and AP Chemistry, honors and AP Physics, Honors Algebra II, Honors Geometry, and Honors Precalculus by preparing lesson plans and notes aligning with the state curriculum, employing engaging examples, and working through practice problems and assignments.
- Mentored students in effective note-taking habits, study skills and plans, time management, and proactive communication in the classroom.

Introduction to Controlled Fusion, Cornell University

Ithaca, NY

Teaching Assistant / Grader (part-time)

Jan 2024 — May 2024

- Held weekly office hours, graded problem sets and exams, and led lectures in the professor's absence for a class of 30 undergraduate and graduate students, communicating advanced physics in clear, simple terms.

SELECTED PROJECTS

ElectroBottle | Innovation, Design & Entrepreneurship, Columbia University

Sep 2025 — Present

- Leading a team of 4 to design a smart water bottle for passive electrolyte and hydration tracking; grounding design in 20+ patient/expert interviews and prototyping 3+ novel electromechanical dispensing systems in SolidWorks.

Rainbow RoboCat | Robotics Studio, Columbia University

Jan 2025 — May 2025

- Owned full-cycle development of a walking quadruped robot on a \$100 budget (Autodesk Inventor, Raspberry Pi, 3D printing, Python); achieved a 38 cm/s gait optimized with ROS and ML-based gait tuning in PyBullet.

CFD Modeling of Arterial Obstruction | Advanced Experimental Physics Lab, Cornell University

Jan 2024 — May 2024

- Simulated blood flow through healthy vs. obstructed arteries in Ansys Fluent (CAD, meshing, boundary conditions); uncovered a 7% peak-wall-stress increase and identified regurgitation regions indicating elevated aneurysm risk.

Atelier Vintage | Independent Sustainable-Fashion Project

Sep 2025 — December 2025

- Founded and built Atelier Vintage, an independent sustainable-fashion project, owning it end-to-end from concept through launch and applying an engineer's systems mindset to a creative product. [Refine with what Atelier Vintage actually makes/does.]

VOLUNTEER & LEADERSHIP

Kiwanis Ambulance Service of Boonton

Boonton, NJ

Youth EMT Squad Member

Mar 2018 — Mar 2020

- Volunteered 250+ hours providing patient care in crews of 3–5 — handling equipment, rig checks, and vital-sign recording — and trained new youth members.

incrAmath

New Jersey

Co-Founder

Mar 2019 — Aug 2020

- Co-founded a free volunteer tutoring organization pairing high schoolers with low-income middle schoolers; built the website, recruited tutors, and designed a sustainable organizational structure.

Girl Scouts of Northern New Jersey

New Jersey

Gold Award Recipient

Sep 2007 — Jun 2020

- Dedicated 80+ hours to a Gold Award project on reducing sports-related injuries, partnering with a physical therapist to research common injuries and build prevention-education resources.